LABORATORY TITLE: INVESTIGATING DJI MAVIC AIR FLIGHT EVIDENCE CASE STUDY GUIDE

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Executive Conclusion

The examination of the provided DJI Mavic Air flight evidence confirms that a flight event occurred on June 19, 2018, and corresponding media artifacts were generated. Flight logs place the aircraft in Parshall, Colorado, with a recorded flight duration of 5 minutes and 5 seconds. However, this investigation is severely constrained by an inability to download the complete raw microSD and EWF image files for examination. This limitation resulted in incomplete cryptographic hashing (Input/output error) and truncated EWF verification. Consequently, while there is moderate confidence that the depicted interfaces and application logs represent genuine flight data, confidence remains low regarding the full binary integrity of the physical media and the specific attribution of the operator.

Evidence Acquisition and Integrity

The following inventory reflects the evidence visible in the provided examination screenshots. Due to incomplete file downloads, full SHA-256 hashing for physical media was not possible**.**

Evidence ID	Original filename	Source/path	Status/limitation
E01	Android Logical Image	Android_Logical.zip / Android_Logical/sdcard/DJI/	
E02	Raw microSD Image	/run/media/kali/2019-1B01/	~8.0 GB FAT32 volume; md5sum failed due to Input/output error indicating incomplete file
E03	EWF Image	DF048.E01	Acquired Jun 28 2018. ewfverify killed

Methodology

The investigation relied on offline visual analysis of the provided screenshots depicting an Android logical extraction and mounted physical media. Mobile application artifacts (dji.go.v4) were identified using terminal commands (tree, ls, stat). Flight telemetry (TXT) was parsed using third-party web interfaces (Flight Reader and Airdata UAV) to interpret coordinates, timelines, and operator names. Media metadata was extracted using mediainfo. The EWF image was mounted via ewfmount, and its file structure was enumerated using The Sleuth Kit (fls) to compare against the raw microSD mount (lsblk, tree).

Investigation Questions

1. How did you preserve the case materials?

Proper preservation via cryptographic hashing (SHA-256) could not be completed. Attempts to hash the raw microSD (md5sum) resulted in Input/output error, and verification of the EWF image (ewfverify) was killed at 26%. This is a direct result of the incomplete file downloads.

2. What DJI GO 4 artefacts are present?

The Android extraction contains the standard dji.go.v4 package directories. Identified artifacts include TXT flight logs in FlightRecord/, DAT telemetry in MCDatFlightRecords/, cached MP4 video in DJI_RECORD/, and thumbnails in CACHE_IMAGE/.

3. What does the main flight record establish?

The parsed TXT record establishes a flight by a Mavic Air on June 19, 2018, starting at 18:50 UTC (12:50 PM Local). The flight took place in Parshall, CO, reaching a max altitude of 93.5 ft and a max distance of 325 ft over 5 minutes and 5 seconds.

4. What does the cached video establish?

The cached video (2018_06_19_14_50_39.mp4) has a duration of 4 minutes and 0 seconds, recorded at a 1280x720 resolution utilizing an AVC codec. Visual inspection shows an outdoor scene with a white tent and vehicles on a grassy plain. The video duration is roughly one minute shorter than the total flight time, indicating the camera was not recording for the entire flight envelope.

5. What does the detailed flight record establish?

A detailed DAT record (18-06-19-02-47-38_FLY057.DAT) was located within the MCDatFlightRecords directory. The file size is 4,002,791 bytes with a modification date of June 19, 2018. While its presence is confirmed, its internal telemetry parameters were not parsed in the provided evidence set.

6. What do cached pictures establish?

Six cached thumbnail images (thumb_...jpg) were identified within the CACHE_IMAGE folder, totaling 95,100 bytes. While these prove images were cached to the mobile device via the application, without EXIF data validation, they cannot independently prove exact geographical coordinates or operator identity.

7. What does the raw internal microSD image show?

The raw image mounts as a 7.9 GiB FAT32 file system under /run/media/kali/2019-1B01. It contains a DCIM/100MEDIA directory housing five high-resolution MP4 files (DJI_0001.MP4 to DJI_0005.MP4) and corresponding .THM files. Visual playback of DJI_0001.MP4 shows an indoor setting featuring a refrigerator, distinct from the outdoor flight event.

8. What does the EWF image show and how does it compare?

The EWF container (DF048.E01) was acquired on June 28, 2018, using FTK Imager. When mounted (ext_sd/ewf1) and parsed using fls, it reveals the exact same 100MEDIA directory structure and file nomenclature (DJI_0001.MP4 through DJI_0005.MP4) as the raw microSD image. This establishes structural equivalence between the two physical media representations.

9. What conclusion can you defend?

It is defensible to state that a DJI Mavic Air linked to the Android device conducted an outdoor flight in Parshall, CO on June 19, 2018, and generated corresponding cached media. However, limitations heavily restrict further conclusions.

Conclusion and Limitations

- Incomplete Evidence Acquisition: The inability to download the full microSD and EWF images is a critical limitation. Hashing failures (Input/output error) and aborted EWF verifications mean binary integrity cannot be established.
- Attribution vs. Account Name: The flight record lists the associated pilot as "cyber guide". This is a digital account moniker and does not technically prove the physical identity of the human operator holding the controller.
- Media Discrepancies: The onboard media (DJI_0001.MP4) shows an indoor scene, while the cached mobile video (2018_06_19_14_50_39.mp4) shows an outdoor tent. This indicates the physical SD card retains historical flights unrelated to the specific flight parsed in the TXT log.

Timeline template

Stage	UTC timestamp	Evidence ID	source	Observed fact	Interpretation and limitation
Device state and power or motor on	2018-06-19 18:50:00	E04	Flight Reader (TXT)	Flight mode changed to Starting Motors (0m 0.2s).	Initiates the active flight sequence; time zone converted from 12:50 PM Local to 6:50 PM UTC
Take off or video start	2018-06-19 18:50:02	E04	Flight Reader (TXT)	Aircraft left the ground logged at 0m 2.3s.	Confirms airborne state; battery at 95%.
Main flight record event	2018-06-19 18:51:31	E04	Flight Reader (TXT)	Maximum flight distance reached warning logged at 1m 31.4s.	Aircraft hit a software boundary (Max Distance: 325 ft).
Landing or video end	2018-06-19 18:55:00	E04	Flight Reader (TXT)	Flight End (UTC) recorded.	Total duration in air: 5m 2s.
Power off and correlation	2018-06-19 18:55:05	E04, E05	Airdata (TXT)	Total Log Duration recorded as 05m 05s.	Flight terminated. Cached MP4 duration of 4m 0s indicates video recording did not span the entire power-on session.

Evidence Log

Evidence ID	Tool & method	Source record/path/image	Observed fact
E01	tree / ls	Android_Logical/sdcard/DJI/	Confirmed presence of DJI GO 4 application structure.
E02	md5sum / lsblk	run/media/kali/2019-1B01/	Identified 7.9 GiB FAT32 volume; hashing failed with I/O error.

E03	ewfinfo / ewfverify	DF048.E01	Identified EWF container; verification failed at 26%.
E04	Flight Reader / Airdata	FlightRecord/ (TXT)	Parsed flight metrics showing Parshall, CO location and "cyber guide" pilot.
E05	mediainfo	DJI_RECORD/2018_06_19_14_50_39.mp4	Extracted video metadata showing 4m 0s duration and AVC codec.
E06	stat	MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT	Located detailed flight telemetry record sized 4,002,791 bytes.
E07	ls	CACHE_IMAGE/	Identified 6 cached picture thumbnails (thumb_...jpg).
E08	fls vs tree	ext_sd/ewf1 vs /run/media/kali/2019-1B01/	Confirmed 100MEDIA structure and MP4 files (DJI_0001.MP4 to DJI_0005.MP4) match across both sources.